Sections: B,C & D

Linear Algebra (MT-1004)

Homework#23

Instructor: Muhammad Ali

Question 1

Consider the dynamical system $X_{k+1} = AX_k$ where $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Classify the origin as an attractor, repeller, saddle point or none of these.

Question 2

Compute e^A for $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$